ENV ENG 264 **Fall 2026**
Due in Parts A, B **Projects: 15%**

Data Analysis Projects

Class project will be done in groups (3-4 in a group) and a formal engineering report (maximum of 4 pages of text and equations excluding abstract, cover page, figures, tables and references) must be submitted individually. Individual project score will count 85% and submitted comments on other team members will count 15%.

Cover Page: Should have project title, your name, names of team-mates (reviewers), and date of submission. The abstract can be included if it fits.

Abstract: Provide a 100-300 word capsule summary of project objectives and important findings. The abstract should stand on its own, and the report should be understandable without the abstract. Write the abstract *after* completing the report.

Introduction: Provide a brief background and full objectives. It should be clear from the introduction why the work and analysis that follows is necessary.

Procedure: Concisely explain the mathematical steps and techniques that will be used. This is designed so that a knowledgeable person can easily grasp what you will be doing without having to get bogged down in the actual calculational details.

Results and Analysis: This is where you exhibit the actual calculations and explain what the results mean, whether they make sense and why. This is the section where figures and tables are presented and discussed. Though R will do the actual number crunching you should describe what is being done. Relegate code to an appendix.

Conclusions: Provide a brief but complete summary of the findings. A busy reader should be able to skip straight from the introduction to the findings and feel satisfied they understood what came from the project. The findings should be presented in greater detail than in the abstract.

References: List useful references and be sure to cite them in the body of the report. It is expected you will do some outside reading to understand the topics fully. Bear in mind that Wikipedia articles are not considered references, but they may be good sources for finding references. The literature is vast, so the once your team finds at least two *useful* sources for each sub project, stop searching.

Presentation Tips

Results are best presented in tables and figures, with text explaining each. All tables and figures should have a number and a caption that explains enough to get the meaning.

All plots should be made in Rstudio. All table columns and plot axes should be properly labeled with units included. Equations should be done with an equation editor, NOT by hand.

Sections should be labelled. ALL tables, figures, plots and equations must be discussed in the text. These discussions should reference the table, equation or figure number. They should be inserted in the text close to where they are referenced. Equations should get a separate line, tables and figures can be placed at top or bottom of the page where they are discussed.

Project Evaluation

	Expectations	**Percentage**
Abstract	Concise, all key points	**5**
Introduction	Clear articulation of the issues being addressed in this project	**10**
Procedure	Someone should be able to read your procedure and then replicate your work with a similar data set.	**25**
Results and Analysis	Present calculations and findings using tables, figures, and explanations. There should be enough detail to find why people get different results.	**30**
Conclusions	The results presented in a way that both technical and non-technical readers can understand them.	**10**
Figures and Tables	All should be clear with labels and captions, and discussed in the text.	**15**
References	Statements in the text that require a reference should have one; each reference should be cited in the text.	**5**

Everyone in the group writes their own paper: so what is the group for? To discuss and review each other's work, offering advice for improvement. A draft version will be submitted to the group folder on blackboard; everyone will submit comments as a filename : authorname_ProjectA_comments_reviewname.doc. Worth 15% of score. Those who do not submit a draft will not get credit for comments they make.

Project Descriptions

Congratulations for being hired for an Environmental Analysis company, which has just won several bids for projects. Your team will be working on two different projects involving an environmental impact assessment of water quality and modeling the urban heat island. You and your team's responsibilities involve analysis for both projects. You must develop a comprehensive report (based on data analysis) on these projects for senior management. You may not find all the information you need in the data sets so will have to do some extra research to pull the project together. State and justify any assumptions you make in the analysis. Your team will consult with each other, but do separate analysis and write-ups to provide cross checks and redundancy to ensure accuracy.

Sub-project A: Water Quality Assessment for Newtown Creek: Oxygen for Fish.
The New York City Department of Environmental Protection (NYCDEP) collects and analyzes data from a total of 85 sampling stations city wide to monitor water conditions and understand how water quality can be improved. They approached your company to analyze the data for the 3 sampling sites on the downstream end of Newtown Creek to evaluate viability for fish. Details of sampling stations and the data are provided here: https://www1.nyc.gov/site/dep/water/harbor-water-quality.page . NYCDEP specifically wants to understand the following:

1) According to New York State Department of Environmental Conservation (NYSDEC) standards a dissolved oxygen level of 5 mg/L or less indicates ecosystem degradation. By this criterion, what is the probability of not meeting the standards at each site in the summers vs winters?
2) What is the probability of not meeting standards simultaneously at all sites? Probability of exactly one of any of the sites not meeting the standard?
3) Are the average DO levels at these sites different from each other at the 95% confidence level?
4) Are daily changes in dissolved oxygen levels at the Newtown Creek sensor sites correlated to each other?

Sub-project B: Manhattan's Urban Heat Island
In the summers of 2012 and 2013 a series of field campaigns to measure Manhattan's urban heat island was conducted by teams of people who would simultaneously walk along designated streets from west to east starting at 2 pm in the afternoon with sensors automatically logging temperature and humidity. In this data set dewpoint (DP) is a measure of water content in the air that does not change with temperature the way relative humidity (RH) does. Data were adjusted for the change in temperature and humidity during the roughly 40 minute walks, and then for each day the data set average was subtracted from the data (referred to as "differences"). To average together data from different days two approaches were used. One was to simply average together the data differences in each location (Tdiff, RHdiff, DPdiff). The other approach was to divide the difference data at each point by the daily standard deviation to form Z scores, and average those (Tdev, RHdev, DPdev). The logged data was averaged together into 100 m segments, and

associated with average surface characteristics within a 100 m square centered on the segment center. These characteristics are:
NDVI – Normalized Difference Vegetation Index: a satellite product that estimates the amount of vegetation. NDVI ranges from -1 to 1, with values greater than 0.1 containing some vegetation, with values above 0.8 being heavy forest.
Albedo – reflectivity of visible light integrated across all wavelengths. Albedo can range from 0 to 1.
Building Area Fraction - Bareafrac: the fraction of the area taken up by buildings
Building Height – Bheight: the average height in meters of buildings in the area
Elevation – the average elevation in meters
Water Fraction – waterfrac: the fraction of open water in the area.

Your task is to come up with multi-variable models for temperature and humidity based on the local surface characteristics. You need to decide whether the data differences or deviations (Z values) are a better fit for temperature, and whether RH or dewpoint produces a better fit for humidity. You may make nonlinear transformations of surface variables, but not of the fitted variables of temperature and humidity/dewpoint. It is suggested that you scale all variables to the same range so that the model coefficients indicate the relative importance of each variable. You should decide which predictor variables to keep and which to discard.

This breaks into two tasks:
• Fit Temperature as a function of the surface variables using either T differences or T deviations and decide which is a better fit. Any surface variable transformations such as scaling should be applied in both cases. Once you have decided whether differences or deviations work best, use this for the next task.
• Fit RH as a function of surface variables and then fit dewpoint as a function of surface variables. Use the same surface variable transformations for both, and decide which is a better fit.

If you are doing nonlinear transformations of variables (such as a logistic transformation) it is a good idea to do the fit first with non-transformed variables, then with transformed variables to see if there is an improvement. This does not apply to scaling, which is a linear transformation so will not affect most measures of goodness of fit.

Write a report that clearly explains the fitting and modifications made to the data in order to create the best fit model. Include all tests of model fitting, and plots to demonstrate how well your model works. It is recommended that you randomly divide the data into training and validation sets, with the larger fraction being the training set. The data is found in **Manhattan_UHI.csv**